

Midterm Solutions for MIAE 221 – Fall 2024 Exam Review Premium

Course: MIAE (221) 261 documents

University: Concordia University

2024/2025

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Engineering and Computer Science

SOLUTIONS

Department of Mechanical, Industrial
and Aerospace Engineering

X-A

MIAE 221 SECTION X – Midterm, Fall 2024

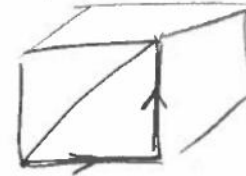
Instructors: Dr. Martin Pugh

Put your answers on the Scantron IBM Sheet. USE PENCIL!!!

Answer all questions. An equation sheet is attached at the end of the exam paper. Questions 1-20 are worth two marks each; Questions 21-30 are one mark each. **Hand-in only the Scantron MC answer sheet – keep the question paper.**

1. The face diagonals of a cubic crystal system are represented by which one of the following family of directions?

- a) $\langle 100 \rangle$
 b) $\langle 110 \rangle$
 c) $\langle 111 \rangle$
 d) $\langle 210 \rangle$
 e) $\{211\}$



2. The activation energy and pre-exponential term (D_0) values for carbon diffusing in BCC – iron are 80 kJ/mol and $6.2 \times 10^{-7} \text{ m}^2/\text{s}$, respectively. The corresponding values for carbon diffusing in FCC – iron are 148 kJ/mol and $2.3 \times 10^{-5} \text{ m}^2/\text{s}$. At 910°C the carbon atoms:

- a) Move ~3 times faster in the BCC iron than in the FCC iron.
 b) Move ~3 times faster in the FCC iron than in the BCC iron.
 c) Move the same speed in both iron structures.
 d) Move ~30 times faster in the BCC iron than in the FCC iron.
 e) Move ~30 times faster in the FCC iron than in the BCC iron.

$$D_{\text{FCC}} = 2.3 \times 10^{-5} e^{-\frac{148000}{8.314 \cdot 1138}}$$

$$D_{\text{BCC}} = 6.2 \times 10^{-7} e^{-\frac{80000}{8.314 \cdot 1138}}$$

3. Diamond and graphite are both composed of carbon atoms, but diamond is much harder and stronger than graphite because:

- a) Diamond contains only covalent bonds whereas graphite contains only van der Waals bonds
 b) Graphite contains only covalent bonds whereas diamond contains only van der Waals bonds
 c) Diamond contains both covalent and van der Waals bonds whereas graphite contains only van der Waals bonds
 d) Graphite contains both covalent and van der Waals bonds whereas diamond contains only van der Waals bonds
 e) Diamond contains only covalent bonds whereas graphite contains both covalent and van der Waals bonds

4. Vanadium has a lattice constant of 0.304 nm. How many vanadium unit cells are there in 1 mm^3 ? (Vanadium is BCC, with an atomic radius of 0.136 nm and a density of 6.1 Mg m^{-3})

- a) 3.56×10^{19}
 b) 3.289×10^9
 c) 3.56×10^{28}
 d) 2.847×10^{20}
 e) 3.289×10^{18}

$$\text{Vol.} = a^3 = (0.304 \times 10^{-6} \text{ m})^3 = 2.8 \times 10^{-20} \text{ m}^3$$

$$\frac{1}{2.8 \times 10^{-20}} = 3.56 \times 10^{19} \text{ atoms}$$

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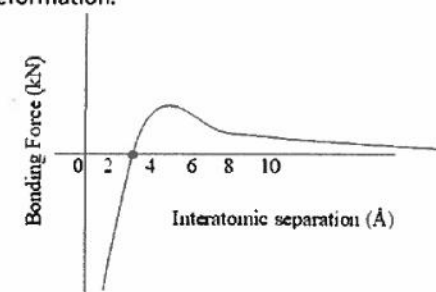
Fall

5. A can of mushroom is dropped from a store shelf onto a concrete floor and is found to be badly distorted out of shape. Which of the following best describes the cause of this distortion?

- a) The tensile strength has been exceeded.
- b) The yield strength has been exceeded.
- c) The elastic modulus of the material has been exceeded.
- d) The mushrooms have spontaneously combusted.
- e) The ductility of the material was insufficient to resist plastic deformation.

6. The accompanying figure represents the net bonding force between two atoms. What is the equilibrium spacing between the two atoms? ($\text{\AA}$ is Angstrom – equivalent 10^{-10} m).

- a) $2 \text{\AA}$
- b) $3 \text{\AA}$
- c) $5 \text{\AA}$
- d) $7 \text{\AA}$
- e) $8 \text{\AA}$

when $F = 0$ 

7. A very thick piece of iron with negligible initial nitrogen concentration is exposed to nitrogen gas at a fixed temperature. The surface concentration of nitrogen is maintained at 5 kg/m^3 . If the diffusivity of nitrogen, at this temperature, is $6.9 \times 10^{-11} \text{ m}^2/\text{s}$, what is the nitrogen concentration 0.5 mm below the surface after 1 hour of exposure? Refer, if necessary, to the error function tables at the end of the exam.

- a) 5 kg/m^3
- b) 0.48 kg/m^3
- c) 1.0 kg/m^3
- d) 2.4 kg/m^3
- e) 6.9 kg/m^3

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = 1 - \text{erf}\left(\frac{0.5 \times 10^{-3}}{2\sqrt{6.9 \times 10^{-11} (3600)}}\right)$$

$$\frac{C_x - 0}{5 - 0} = \frac{C_x}{5} = 1 - \text{erf}(0.5016) = 1 - 0.5205$$

$$C_x = 2.4 \text{ kg/m}^3$$

8. If a smooth block of copper is pressed against a smooth block of nickel and both are heated to a high temperature (but below their melting points), and held for a long time:

- a) Copper atoms will diffuse into nickel because copper has the lower melting point.
- b) Nickel atoms will diffuse into copper because copper has the lower melting point.
- c) Copper atoms will diffuse into the nickel and nickel atoms will diffuse into the copper.
- d) No diffusion will occur between these metals because they have complete solid solubility in each other.
- e) Nickel will diffuse into copper because copper has a lower packing factor and more empty space available for the nickel atoms.

9. Consider two types of "Low carbon" steel, Steel A and Steel B. Steel A has twice the yield strength of B but only half the ductility. If an attempt is made, at room temperature, to elastically bend identically sized bars (3 metre long, $25 \times 25 \text{ mm}$ square cross-section) by the same amount for each type of steel:

- a) A would be easier (ie require less force) to bend initially than B
- b) B would be easier to bend initially than A.
- c) There would be no difference in the forces required to bend the bars initially
- d) There would be a difference in the forces required to bring about initial elastic bending, the difference being a function of the yield strengths of the bars rather than the tensile strengths.
- e) This cannot be done because the two steels have different ductility

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10. A self-interstitial is

- a) a small atom that is
- b) an empty lattice po
- c) an extra-large inter
- d) an atom from the crystal that is crowded into an interstitial space between similar atoms

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10. A self-interstitial is

- a) a small atom that is crowded into the interstitial spaces of the lattice of larger atoms
- b) an empty lattice position
- c) an extra-large interstitial space
- d) an atom from the crystal that is crowded into an interstitial space between similar atoms
- e) an atom that likes to take pictures of itself

11. Steel rods, 20mm in diameter, are used to suspend an aerial walkway. The rods are each 10m long prior to installation. After construction the rods each carry a load of 1000 kg (9800 N). The stress is below the yield strength of the steel. The Young's modulus of the steel is 207 GPa. The length of a rod after construction is now:

- a) 10.00151 m
- b) 10.000338 m
- c) 10.151 m
- d) 10.00015 m
- e) 10.00227 m

$$\epsilon = \frac{\Delta L}{L_0} = \frac{\sigma}{E} = \frac{F}{AE} = \frac{F}{\pi d^2 E} = \frac{4 \cdot 9800}{\pi (20 \times 10^{-3})^2 \cdot 207 \times 10^9}$$

$$\Delta L = 1.507 \times 10^{-3} \text{ m}$$

$$L_f = 10.00151 \text{ m}$$

12. If the walkway in the above question is now occupied with people who increase the load on each rod by 50%, the new length of the rods is:

- a) 10.00151 m
- b) 10.0023 m
- c) 10.32 m
- d) 12.27 m
- e) 1000.151 mm

$$\Delta L_2 = 1.5 \Delta L_1 = 2.26 \times 10^{-3} \text{ m}$$

$$L_2 = 10 + 0.00226 = 10.0023 \text{ m}$$

13. In the adjacent Figure, Plane K is best described by the following Miller indices:

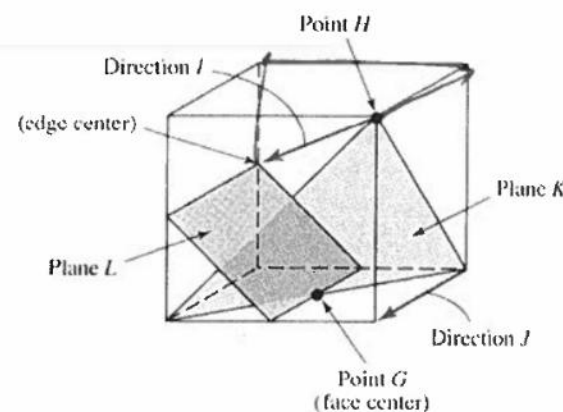
- a) (111)
- b) (100)
- c) [111]
- d) $[\bar{1}\bar{1}0]$
- e) $(\bar{1}\bar{1}1)$

$$\begin{array}{ccc} x & y & z \\ -1 & -1 & 1 \\ -1/1 & -1/1 & 1/1 \\ (\bar{1} \bar{1} 1) \end{array}$$

14. In this Figure, Direction I is best described by the following Miller indices:

- a) [122]
- b) [221]
- c) $(\frac{1}{2} \frac{1}{2} 0)$
- d) $[0\bar{2}\bar{1}]$
- e) $[\bar{2}\bar{2}\bar{1}]$

$$\begin{array}{ccc} x & y & z \\ -1 & -1 & -1/2 \\ -2 & -2 & -1 \\ [\bar{2} \bar{2} \bar{1}] \end{array}$$



15. The expected diffraction angle (2θ) for the first-order reflection from the (113) set of planes for FCC platinum (atomic radius of platinum is 0.1387nm) when monochromatic radiation of wavelength 0.1542 nm is used is:

- a) 5°

$$n\lambda = 2d \sin \theta \quad d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad a = 2R\sqrt{2}$$

$$\sin \theta = \frac{n\lambda}{2d}$$

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15. The expected diffraction angle (2θ) for the first-order reflection from the (113) set of planes for FCC platinum (atomic radius of platinum is 0.1387nm) when monochromatic radiation of wavelength 0.1542 nm is used is:

a) 5°

$$n\lambda = 2d \sin \theta \quad d = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad a = 2R\sqrt{2}$$

$$\sin \theta = \frac{n\lambda}{2d}$$

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b) 29° c) 63° → ☒ d) 81° e) 180°

$$d = \frac{2(0.1387 \times 10^{-9})\sqrt{2}}{\sqrt{1^2 + 1^2 + 3^2}} = 1.183 \times 10^{-10}$$

$$\sin \theta = 0.652 \quad \theta = 40.68$$

$$2\theta = 81.3$$

16. Which of the following best describes covalent bonding?

a) One atom donates its outer electron(s) to a different atom and they are electrostatically attracted.

→ ☒ b) One atom shares its outer electron(s) with a neighbouring atom(s) and both atoms become energetically more stable.

c) Hydrogen molecules join the atoms together.

d) The atoms plastically deform and necking occurs between them making them stick to each other.

e) One atom forms a random dipole and attracts an atom with an induced dipole.

17. The density of thorium, which has the FCC structure, is 11.72 g/cm³. The atomic weight of thorium is 232 g/mol. What is the lattice parameter, a , of thorium?

a) Thorium is a radioactive metal and as such does not form crystals

→ ☒ b) 5.0856×10^{-8} cmc) 2.1487×10^{-8} cmd) 10.0943×10^{-8} cme) 15 Å (1 Å = 1×10^{-10} m)

$$\rho = \frac{nA}{V_c N_A}$$

$$a = \sqrt[3]{\frac{nA}{\rho N_A}} = \sqrt[3]{\frac{4 \times 232}{11.72 \times 6.022 \times 10^{23}}} = 50.8 \times 10^{-9} \text{ cm}$$

18. A steel gear is to be carburized (using a carbon-rich gas at the surface to diffuse carbon atoms into the steel). It has been determined that the current procedure takes too long. What can be done to decrease the time required to obtain the same carbon concentration?

a) use a steel with a lower initial carbon content (C_0) so that the concentration gradient is higher.b) increase the temperature, T (without changing the crystal structure)c) increase the concentration of carbon in the surface gas (C_s).

d) use a steel that has been stabilized in the FCC structure rather than BCC.

→ ☒ e) both b and c.

19. During a tensile test of a cylindrical brass rod with an initial gauge length of 50 mm and diameter of 10 mm, an increase of 0.01 mm in length and a decrease in diameter of 6×10^{-4} mm is observed. What is the Poisson's ratio of this specimen?

a) 0.35

→ ☒ b) 0.30

c) 0.25

d) 0.4

e) There is not enough information to find the Poisson's ratio.

$$\nu = -\frac{\epsilon_x}{\epsilon_z}$$

$$\frac{\Delta L}{L_0} = \frac{0.01}{50} = 2 \times 10^{-4}$$

$$\frac{\Delta D}{D_0} = \frac{6 \times 10^{-4}}{10} = 6 \times 10^{-5}$$

$$\nu = -0.30$$

20. Which of the following mechanical properties cannot be measured from a tensile test?

→ ☒ a) Hardness

(b) Yield strength

(c) Tensile strength

(d) Young's Modulus

(e) Ductility

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True or False: For Questions 21 – 30, decide whether the following statements are True or False and shade in the appropriate box on the Scantron Answer sheet. Each correct answer is worth 1 mark.

For True, shade in letter (a). For False shade in letter (b).

- F 21. The plastic region of the stress-strain curve for a metal is characterized by a linear relationship between stress and strain.
- T 22. Generally, a weakly atomically bonded solid will have a lower melting point, lower elastic modulus, and higher thermal expansion coefficient compared to a strongly bonded atomic solid.
- F 23. Ductility is the ability of materials to absorb energy during plastic deformation and then give it back when unloaded.
- T 24. The reason why ice floats on water is due to the lower density of the ice structure which is a result of the secondary (hydrogen) bonding that forms between H₂O molecules during freezing.
- F 25. Amorphous or glassy atomic structures have both long range and short range order whereas crystalline atomic structures have only long-range order.
- T 26. The face centered cubic system has the highest packing factor in the cubic system.
- F 27. The interstitial diffusion coefficient for a small atom diffusing in a metal with larger atoms is usually smaller (slower) than the diffusion coefficient for an atom that diffuses in that same metal via vacancy diffusion because more energy is needed to squeeze the interstitial atoms between the larger atoms.
- T 28. The Young's Modulus of a solid metal is strongly related to the strength of the bonds between atoms and so it is not possible to significantly change the Young's Modulus of this metal unless bonds are changed e.g. through alloying.
- F 29. "Polycrystalline" refers to the elements (eg. carbon) that can exist in more than one crystal form depending on temperature and pressure, whereas "single crystal" refers to elements, such as copper, that only have one possible crystal structure.
- F 30. BCC has 2 atoms/unit cell, FCC has 4 atoms/unit cell and HCP has 3 atoms/unit cell.

EQUATION SHEET ON NEXT PAGE

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MIAE 221 – Midterm Equation Sheet $A \cong Z + N$

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$$\int_r^\infty F_N dr = E_A + E_R$$

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MECH 221 – Midterm Equation Sheet

$$A \cong Z + N$$

$$F_N = F_A + F_R$$

$$E_N = \int_r^\infty F_N dr = E_A + E_R$$

$$\%ionicity = \{1 - \exp[-(0.25)(X_A - X_B)^2]\} \times 100$$

$$a = 2R\sqrt{2}$$

$$a = \frac{4R}{\sqrt{3}}$$

$$APF = \frac{\text{volume of atoms in unit cell}}{\text{total unit cell volume}}$$

$$\rho = \frac{nA}{V_C N_A}$$

$$LD = \frac{n_a}{L_l}$$

$$PD = \frac{n_a}{A_p}$$

$$n = 2^{G-1}$$

$$n\lambda = 2d_{hkl} \sin \theta$$

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$

$$J = \frac{M}{At}$$

$$J = -D \frac{dC}{dx}$$

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

$$\sigma_{Eng} = \frac{F}{A_0}$$

$$\epsilon_{Eng} = \frac{\Delta l}{l_0}$$

$$\sigma = E\epsilon$$

$$\tau = G\gamma$$

$$\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$$

$$E = 2G(1 + \nu)$$

$$\%El = \left(\frac{l_f - l_0}{l_0}\right) \times 100$$

$$\%RA = \left(\frac{A_0 - A_f}{A_0}\right) \times 100$$

$$U_r = \frac{1}{2} \sigma_y \epsilon_y = \frac{\sigma_y^2}{2E}$$

Constants

$$N_A = 6.022 \times 10^{23} \text{ atoms/mol}$$

$$k = 8.62 \times 10^{-5} \text{ eV/atom.K or } 1.38 \times 10^{-23} \text{ J/atom.K}$$

$$R = 8.314 \text{ J/mol.K}$$

z	erf(z)	z	erf(z)	z	erf(z)
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.00	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

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